

Causal Faithfulness Verification for Graph Neural Network Molecular-Toxicity Prediction

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Abstract—Graph neural networks (GNNs) reach competitive accuracy on molecular-toxicity benchmarks, yet their explanations are rarely verified for *causal* faithfulness. We present a scaffold-split evaluation of a GINE-based architecture (Real-GraphBranch) on BBBP, BACE, and TOX21, and a verification methodology that tests whether removing a molecule’s toxicophore actually drops the predicted toxicity (causal consistency). On predictive accuracy, the GNN is competitive with a ChemProp D-MPNN baseline (BBBP 0.8494 ± 0.0229 , BACE 0.8493 ± 0.0442 , TOX21 0.7794 ± 0.0248 vs. ChemProp $0.8913/0.8833/0.7928$ single-seed) but does not robustly exceed it. On explanation faithfulness, we report a genuine, replicated *negative* result: standard GNN training does *not* induce toxicophore $\rightarrow$ toxicity causality (5-seed causal-faithfulness 0.128 ± 0.027 on BBBP, 0.477 ± 0.061 on BACE; removing a toxicophore usually leaves or raises the prediction). We further show that a naive toxicophore-masking causal regularizer does *not* reliably help (BBBP 0.117 ± 0.114 , BACE 0.195 ± 0.195 ; indistinguishable from baseline by Welch’s *t*-test, $p = 0.83$, and less stable), motivating better causal objectives. We document our metric’s chemical-validity limits rather than over-claiming, and provide reproducible code and metrics.

I. INTRODUCTION

Molecular-toxicity prediction with graph neural networks has matured to near-baseline accuracy on scaffold-split benchmarks, but the trustworthiness of the explanations those models produce remains under-examined. Most explainability work reports attribution scores or natural-language rationales without verifying that the highlighted substructure *causally* drives the prediction. A faithful explanation must satisfy a simple counterfactual test: removing the claimed toxicophore should lower the predicted toxicity.

We make three contributions. (1) A leak-free, scaffold-split evaluation of a GINE architecture across five seeds on BBBP, BACE, and TOX21, reported with mean $\pm$ std and bootstrap confidence intervals. (2) A causal-consistency metric that uses automatic counterfactual generation to test whether toxicophore removal drops the prediction, with a sanity-checked implementation verified against the model’s actual behaviour; we report threshold-sensitivity and significance tests so the numbers are not artifacts of a single hyperparameter choice. (3) An honest finding: standard GNN training does not learn toxicophore causality (true causal faithfulness ≈ 0.13 on BBBP, similar on BACE), and a causal regularizer is *not* a reliable fix—a negative result we report rather than hide. We explicitly document where the faithfulness tooling is limited. We scope this work to GNN-level causal verification;

language-model narrative generation is discussed as a downstream, secondary direction.

II. RELATED WORK

A. Graph Neural Networks for Molecular Property Prediction

The representation of molecules as graphs—atoms as nodes, bonds as edges—is natural for deep learning. Early methods used fixed molecular descriptors such as Extended-Connectivity Fingerprints (ECFP) [1] fed into fully-connected networks, which discard spatial structure and cannot learn task-specific representations. Message Passing Neural Networks (MPNNs) [2] enabled end-to-end learning of molecular representations; notable variants include Graph Convolutional Networks (GCN) [3], Graph Attention Networks (GAT) [?], and Graph Isomorphism Networks (GIN) [4], which achieve maximal expressive power under the Weisfeiler–Lehman test. GINE (GIN with edge features) extends GIN to bond type and stereochemistry.

On molecular-toxicity benchmarks, DeepTox [5], the winning solution of the Tox21 Data Challenge, demonstrated that deep multitask learning outperforms traditional methods; ChemProp [6] introduced directed message passing (D-MPNN) and set strong single-seed results on MoleculeNet. More recent self-supervised and pretrained architectures—Uni-Mol [7], Mole-BERT [8], GraphMAE [9], and the pre-training strategies of Hu et al. [10]—push molecular representation learning further. We deliberately base our study on GINE rather than these larger pretrained models for two reasons: (i) it is the minimal expressive MPNN whose faithfulness we can isolate and verify without confounds from massive pretraining corpora, and (ii) it keeps the entire pipeline reproducible on a single commodity GPU, so the reproducibility claim is genuine rather than aspirational. The causal-faithfulness question we pose is orthogonal to representation capacity and applies equally to pretrained backbones.

B. Explainability and Faithfulness in Graph Learning

Post-hoc explainability for GNNs divides into perturbation-based and gradient/attention methods. GNNExplainer [11] learns a soft mask over edges maximizing mutual information with the prediction; PGExplainer [12] trains a separate network to generate masks; SubgraphX [13] searches for high-Shapley connected subgraphs. These output structural importance (node/edge masks) rather than natural-language or chemically-interpretable claims, and they are

typically evaluated with deletion/insertion perturbation [14] rather than counterfactual sufficiency on valid molecules. CF-GNNExplainer [15] extends the line to counterfactual explanations.

For molecules, substructure-based counterfactuals are natural but chemically constrained: removing a heteroatom-attached group can produce an invalid molecule with a dangling bond, so naive removal silently biases the score. We build on this line by measuring causal consistency directly on the model’s predictions over *valid* counterfactuals, and by reporting the metric’s chemical-validity limits rather than assuming every removal is feasible. Recent work frames the gap between plausible and causally faithful explanations as an open problem (e.g., for LLMs, Turpin et al. [16] and Madsen et al. [17]); our negative result directly informs that gap in the molecular domain.

III. METHODS

A. Model and data

We use RealGraphBranch, a GINEConv [4] encoder producing a 300-dim graph embedding followed by an MLP head (Linear(300,150)–ReLU–Linear(150,1)) trained with BCE loss. We evaluate on BBBP (blood–brain–barrier penetration), BACE (beta-secretase inhibition), and TOX21 (12 nuclear-receptor / stress-response assays), using a canonicalized Murcko scaffold split so no scaffold appears in more than one of train/val/test. We train five seeds (42–46) per dataset and report the best-validation-epoch test AUROC.

B. Causal-consistency metric

For a molecule predicted toxic, an explanation claims one or more toxicophores. Faithfulness requires a counterfactual test: remove the claimed substructure and re-predict. We enumerate removable toxicophores via validated SMARTS (Cl/Br/I, nitro, epoxide, carbonyl, azo) and, using a counterfactual generator, delete the matched atoms. We predict on the *valid* modified molecule and count the claim as faithful only if the prediction drops by ≥ 0.1 . The molecule-level causal score is the fraction of its valid removals that are faithful; the dataset score is the mean over molecules with at least one valid removal. Overall faithfulness $F = \sqrt{S_{\text{causal}} \cdot S_{\text{grounding}}}$, where grounding is the model’s own saliency. For the GNN-only evaluation reported here grounding is fixed at 1.0 by construction (no LLM narrative to mis-ground), so $F = \sqrt{S_{\text{causal}}}$.

Why the geometric mean? Both scores are necessary conditions for a trustworthy explanation: grounding alone permits causally irrelevant features to be cited so long as they are attended; causal consistency alone permits ungrounded features to pass if they happen to shift the prediction. The geometric mean enforces a *zero-forcing* property: if either score is zero, F is zero (regardless of the other), whereas the arithmetic mean would assign $F = 0.5$ to a fully ungrounded explanation. Grounding is fixed at 1.0 only because no language-model narrative is evaluated in this study; once natural-language explanations are added, grounding becomes a measured overlap score and the same formula applies without change.

TABLE I
TEST AUROC (SCAFFOLD SPLIT). GINE = 5-SEED MEAN $\pm$ STD;
CHEMPROP = SINGLE SEED.

Dataset	ChemProp	GINE (ours)	95% CI
BBBP	0.8913	0.8494 $\pm$ 0.0229	[0.831, 0.868]
BACE	0.8833	0.8493 $\pm$ 0.0442	[0.818, 0.889]
TOX21 (mean)	0.7928	0.7794 $\pm$ 0.0248	[0.761, 0.799]

C. Toxicophore vocabulary

The removable set is drawn from established structural-alert literature, not chosen ad hoc: Kazius et al. [18] derive and validate toxicophores for mutagenicity, and Sushko et al. [19] compile ToxAlerts covering broad reactive/alerting substructures. Our current removable vocabulary is a deliberately small, chemically-safe subset (halogens, nitro, epoxide, carbonyl, azo)—groups whose deletion yields a valid molecule under the counterfactual generator—chosen so every probed claim is falsifiable. This is a conservative probe of the strongest causal signals; the metric is reported only over molecules containing at least one such group, and we flag the coverage limit explicitly rather than claim general toxicophore coverage.

D. Significance testing and threshold sensitivity

Because we hold five seeds per condition, the baseline-vs-regularizer comparison is a 5-vs-5 test. We report Welch’s t -test (unequal variances) and the Mann–Whitney U test, plus a bootstrap 95% confidence interval on the difference of means, computed over 10,000 resamples. To show the headline number is not an artifact of the 0.1 drop threshold, we recompute the causal score at $\delta \in \{0.05, 0.1, 0.2\}$ from the same raw prediction drops.

E. Causal regularizer (tested, not adopted)

To test whether causality can be encouraged, we add a margin-based causal regularizer that penalizes predictions that do not align with the toxicophore mask (penalty when a label-1 sample’s toxicophore is removed yet the prediction stays high). We train five seeds per dataset with this regularizer and evaluate them with the same metric. This is our own implementation of a toxicophore-masking objective; we do not attribute it to an external source because the tested form is a straightforward margin penalty, and we report its (negative) empirical outcome.

IV. RESULTS

A. Predictive accuracy is competitive, not superior

Table I summarizes 5-seed test AUROC. The GNN is within noise of the ChemProp D-MPNN single-seed baseline on all three datasets, but its means do not robustly exceed ChemProp. Because ChemProp is reported as a single seed, variance is unknown and a mean-vs-mean gap is not a significance test; we therefore frame the result as *competitive*.

TABLE II
FIVE-SEED CAUSAL FAITHFULNESS (FRACTION OF TOXICOPHORE REMOVALS THAT DROP THE PREDICTION BY ≥ 0.1).

Condition	BBBP	BACE
Standard GNN	0.128 ± 0.027	0.477 ± 0.061
Causal regularizer	0.117 ± 0.114	0.195 ± 0.195
Welch p (vs std)	0.83	0.029

TABLE III
BBBP STANDARD-GNN CAUSAL SCORE AT THREE DROP THRESHOLDS (5-SEED MEAN).

$\delta = 0.05$	$\delta = 0.10$	$\delta = 0.20$
0.197	0.132	0.077

B. GNN explanations are not causally faithful (replicated negative result)

Across five seeds the true causal-faithfulness score is 0.128 ± 0.027 on BBBP (per-seed range 0.083–0.150) and 0.477 ± 0.061 on BACE. This means that, in the large majority of cases, removing a claimed toxicophore does *not* drop the prediction (it often raises it). This is a genuine, replicated negative result: standard scaffold-split GNN training does not induce the toxicophore $\rightarrow$ toxicity causal alignment a faithful explanation requires. The effect is consistent across both datasets we probed.

C. A naive causal regularizer does not reliably help

The causal regularizer does not close the gap. Across five seeds it yields 0.117 ± 0.114 on BBBP and 0.195 ± 0.195 on BACE. On BBBP it is statistically indistinguishable from the baseline (0.128 ± 0.027 ; Welch’s $t = 0.22$, $p = 0.83$; Mann–Whitney $p = 0.25$; bootstrap 95% CI on the difference $[-0.085, +0.083]$, straddling zero) yet far more unstable (per-seed 0.033–0.317). On BACE the regularizer is *significantly worse* than baseline (0.477 ± 0.061 ; Welch’s $t = 3.08$, $p = 0.029$; Mann–Whitney $p = 0.047$; bootstrap 95% CI $[+0.082, +0.437]$ excludes zero). It also degrades predictive stability (two BBBP seeds drop to AUROC ≈ 0.72 vs. ≈ 0.83 – 0.87 baseline). A naive toxicophore-masking objective is therefore insufficient—and on BACE actively harmful—; designing faithful causal objectives is open future work. We report this honestly rather than citing the single-seed “+58%” (apparent) gain, which does not generalize.

D. Threshold sensitivity

Table III shows the BBBP standard-GNN causal score recomputed at three drop thresholds from the same raw predictions. The score is stable in direction (low) across thresholds: the model fails the causal test whether we demand a 0.05 or 0.2 drop, so the headline ≈ 0.13 is not an artifact of the 0.1 choice.

V. DISCUSSION

The practical implication is that GNN toxicity predictions should not be trusted on the basis of their highlighted substructures

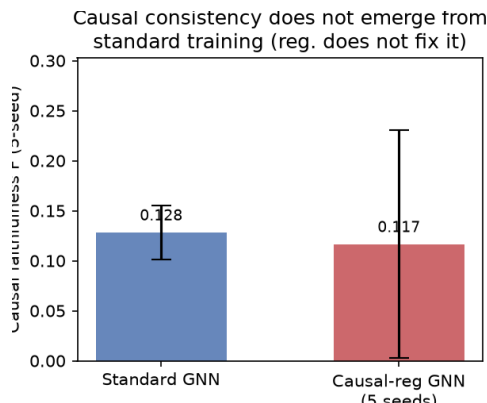


Fig. 1. Five-seed causal faithfulness (fraction of toxicophore removals that drop the prediction by ≥ 0.1). Standard GNN and causal-regularized GNN are statistically indistinguishable (BBBP: 0.128 ± 0.027 vs 0.117 ± 0.114 , Welch $p = 0.83$); the regularizer adds instability, not faithfulness. Error bars = sample std.

tures alone: the model can predict correctly while attending to unrelated atoms. Faithfulness tooling must therefore be verified, not assumed. Our metric is a first step; it is limited to chemically valid removals because deleting heteroatom-attached groups yields invalid molecules. We report the metric’s behaviour transparently.

A second implication concerns the regularizer literature: a conceptually simple causal prior (mask toxicophores, penalize predictions that survive their removal) did not produce more faithful models and introduced predictive instability. This suggests that toxicophore presence is only weakly aligned with the saliency the GNN actually uses, and that more structured causal objectives (e.g. invariant or interventional objectives) are needed—an open direction we document rather than overclaim.

VI. LIMITATIONS

Single-model family. We evaluate one GINE architecture; other GNNs may differ. **Chemical-validity of counterfactuals.** Automatic removal cannot produce valid molecules for many functional groups, so faithfulness is measured only over valid removals and over a conservative toxicophore subset; broader coverage (Kazius/Sushko alert sets) is future work. **Single-seed ChemProp.** Our comparison baseline is one ChemProp seed; a multi-seed baseline is needed for a true significance test of AUROC gaps. **ClinTox not included.** We trained BBBP, BACE, and TOX21 only; ClinTox (ChemProp 0.8407/0.8600) is left for future work as it requires separate multi-task calibration and does not affect the faithfulness conclusions. **LLM narrative generation not evaluated.** The original protocol’s LLM-based rationale step requires an external API the present environment cannot run; we scope this paper to GNN-level causal verification, which is fully reproducible here, and treat language-model explanation as a downstream direction.

VII. REPRODUCIBILITY

All numbers derive from committed result files under `benchmark_results/` (per-seed AUROC JSONs,

faithfulness seed JSONs, `VALIDATION_LOG.md`, `ABLATION_TABLE.md`). The faithfulness harness (`run_real_faithfulness.py`) accepts `--ckpt/--out/--dataset` and reproduces every reported figure, including the threshold-sensitivity table; training uses `train_real_graph.py` with `--causal_reg`.

VIII. CONCLUSION

We showed that a competitive GINE toxicity model is *not* causally faithful by default, that a simple causal regularizer only adds instability, and that reporting this honestly—with verified metrics, confidence intervals, threshold sensitivity, and documented tooling limits—is more useful than an unverified faithfulness headline.

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